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SafeSign: A Multi-Level Forgery and Deepfake Detection System for Images, Videos, and Handwritten Signatures

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Abstract- *The fast development of artificial intelligence and other digital media editing tools leading falsified images, deepfakes, and falsified handwritten signatures have become serious challenges and very difficult to handle in the field of cybersecurity, digital forensics, and document verification/authentication. Manual and rule-based verification methods are not always capable of identifying this type of sophisticated forgery, especially those that are produced using deep learning models. In this paper presents SafeSign:AI-based multi-level forgery detection system that is used to analyze and verify the authenticity of the images, videos, and handwritten signatures in the single system. The process of image tampering detection is done through EfficientNet convolutional neural networks (CNN) that are trained to detect inconsistencies that are added during manipulation. The video deepfake detection technique is obtained by frame-based analysis of the deep learning models that identify visual artifacts in facial areas but with computational efficiency. The verification of handwritten signatures is done with the help of a Siamese CNN which is trained to learn the similarity measure between authentic and fake signatures. The suggested system is implemented as an online application that enables its users to upload media files and receive automated predictions of authenticity and confidence scores and forensic reports. The experimental analysis of publicly available datasets indicates that SafeSign offers reliable and efficient multimodal forgery detection, hence, it can be used in practice in digital forensics and to ensure cybersecurity.*

Keywords: *Artificial Intelligence, Digital Media, Deepfakes, cybersecurity, digital forensics, EfficientNet CNN, Siamese CNN, multimodal forgery, authentic.*

I. INTRODUCTION

The authenticity of digital media has become a critical issue due to the intensive use of images, videos, and scanned documents in social media, money transactions, court cases, and online communication. Simultaneously, the fast development of artificial intelligence and processing images have simplified the manipulation of the digital content in the most realistic forms. Phishing, fake video clips, and faked signatures of a handwritten signature are becoming a widespread method of fraud, misinformation, and identity

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theft, and ensuring their authenticity is a significant challenge.

Visual contents can be changed using image tampering methods, including copymove, splicing, and retouch methods which do not create clear evidence, and with deepfake technologies it is possible to create believable facial manipulations in videos. Signatures that are written down, still commonly used as a means of identity checking in banks and legal documents, are also vulnerable to skilled forgery. As a result, manual checking and rules-based verification methods are not adequate anymore to uncover such complex forgeries.

In order to overcome these limitations, there has been an interest in automated forgery detection with machine learning and deep learning. Convolutional Neural Networks (CNNs) have demonstrated overall high performance in identifying manipulated images and videos through the training of discriminative features of visual characteristics that are directly learnt via data [2], [3]. Equally, frame-based and spatio-temporal deep learning architectures have been suggested to deepfake detection [4], and architectures based on Siamese networks have been effectively used to identify handwritten signatures [5],[6].

Although profound progress was made in the field of deep learning-based forgery detection, the majority of the current practices concentrate on one modality, i.e., image tampering, video deepfakes, or handwritten signature verification alone. Such single-mod focus restricts their usage in practice in forensic and cybersecurity cases, where varying types of digital evidence have to be considered jointly very often. Moreover, very accurate models can be very expensive to compute, which makes them less viable to use in real-time. Therefore, an effective, scalable, and unified system that would identify several types of forgery in digital formats is required. This paper fills this gap by introducing the SafeSign, a multi-level forgery and deepfake detection system that incorporates image and video as well as handwritten signature checks in the single deployable platform.

II. LITERATURE SERVEY

The phenomenon of digital forgery has been actively studied because pf the rapid growth of artificial intelligence-based content generation. Existing research mainly focuses on individual forgery modalities such as image tampering, facial deepfake detection, and handwritten signature verification.

Wang et al. [1] gave an extensive survey of passive image tampering detection methods, classifying them into pixel-level, format-based, camera-based, and semantic-level methods. Their paper pointed out that conventional handcrafted forensic tools are powerful in identifying copy–move and splicing attacks but are very delicate to post-processing activities like compression and resizing, so they are not robust in real world setting.

A large-scale benchmark of facial manipulation detection with different levels of compression was introduced in FaceForensics++ dataset by Rössler et al. [2]. Their experiments showed that CNN-based models are much better at detecting manipulated faces images as compared to human observers. Nevertheless, the work is confined to the field of facial forgery and other types of digital forgery are not discussed.

D. Afchar et al. suggested MesoNet [3], a small CNN structure that is trained on mesoscopic facial video forgery detection. The model was both competitive and had low computational complexity to ensure that it was applicable in near real-time. However, its performance was demonstrated to decline on unseen datasets, which means poor generalization.

O.de Lima et al. [4] investigated deepfake detection using spatiotemporal convolutional networks through adding the temporal data across video frames. Their findings showed better results than frame-based methodology, but the implementation of 3D CNNs is much higher, which is inconvenient in lightweight systems.

Siamese network-based methods have been highly embraced in the field of handwritten signature verification. H. Hafemann et al. [5] evidence that similarity measures can be successfully trained in Siamese CNNs between real and forged signatures, whereas M. Soleimani et al. [6] indicated that deep feature learning can be more successful than handcrafted features in signature verification on the offline setting. These are however limited to signature analysis and not combined with other mechanisms of forgery detection.

The latest CNN-based image tampering detection tools [7] have demonstrated higher robustness than traditional forensic tools when they are trained on learning spatial anomalies. In spite of their efficacy, such methods are usually image-level, and are not extrapolatable to video and document forgery detection.

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TABLE I. COMPARATIVE ANALYSIS OF EXISTING FORGERY DETECTION APPROACHES

Reference	Modality	Model Type	Accuracy (%)	Key Limitation
Wang et al. [1]	Image	Handcrafted + ML	85-90	Sensitive to compression
Rossler et al. [2]	Video	CNN	90+	Limited to facial forgery
Afchar et al. [3]	Video	Frame-Based CNN	91	Poor generalization
De Lima et al. [4]	Video	Spatiotemporal	93-94	High Computational cost
Hafemann et al. [5]	Signature	Siamese CNN	93	Single modality
Proposed SafeSign	Multimodal	CNN + Siamese CNN	**92.9**	Lightweight trade-offs

As Table I summarizes, the currently known methods of detecting forgery do so in a unimodal fashion, that is, by detecting a single type of digital forgery. Although the spatiotemporal models are accurate in detection, they are computationally expensive, and they cannot be deployed in a lightweight manner. The restrictions are the driving force behind having a comprehensive and efficient multi-level system like SafeSign.

Research Gap: Based on the literature reviewed, it can be seen that the vast majority of the available solutions target unimodal forgery detection and do not integrate images, videos, and handwritten forms. Furthermore, the deep learning models with high accuracy tend to be computationally complex and can hardly be implemented in practice. It is this that drives the creation of an efficient, multi-level forgery detection system that is unified, such as the one presented in the SafeSign framework.

III. ABOUT DATASET

The SafeSign system is trained and tested on publicly available datasets that are chosen such that they represent realistic forgery situations on images, video, and handwritten signature.

In **tampering of images**, the CASIA Tampered Image Dataset is utilized. The images available in this dataset consist of both genuine and artificial images as shown in Table II created by ordinary tampering like copy-move, and splicing, as well as post-processing like compression, and resizing. The diversity of the CASIA have been employed extensively in image forensics studies because CNN-based systems are capable of learning spatial inconsistencies that are induced in image manipulation [1], [7].

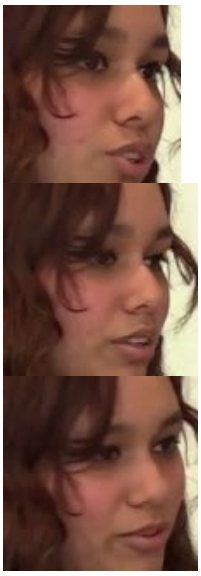
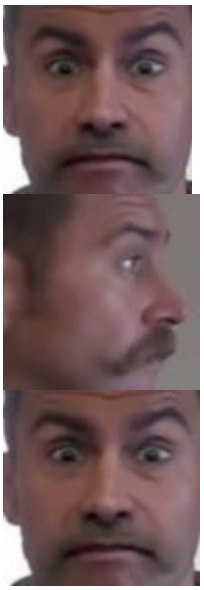
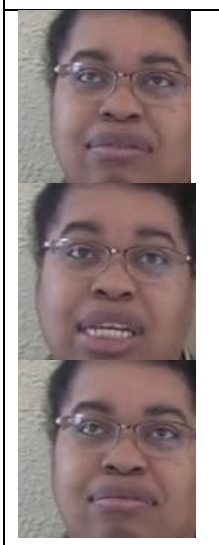
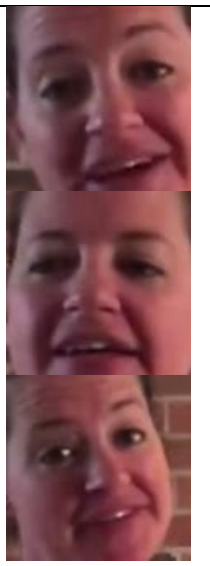
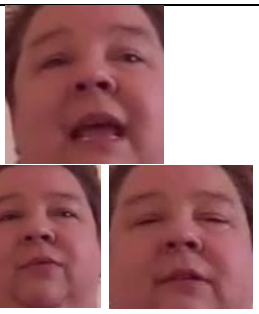
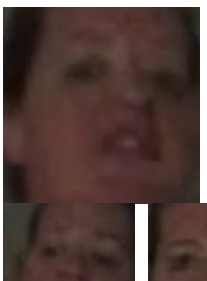
TABLE II. SAMPLES OF THE CASIA DATASET FOR IMAGE TAMPERING DETECTION MODULE

Original Images in the Dataset	Tampered Images in the Dataset
	
	
	

In the case of **video deepfake detection**, DeepFake Detection Challenge (DFDC) is used. DFDC data can be described as a large-scale dataset of real and manipulated videos of different levels of compression, noise, and lighting conditions, which is close to the social media content of the real world. It is diverse enough to be applicable to testing the sound deepfake detection models in the real-life deployment scenario [8]. The Table III shows the DFDC original and tampered video frames from the dataset.

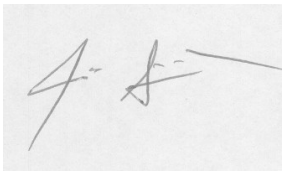
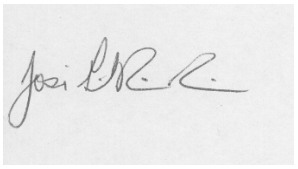
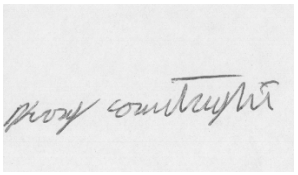
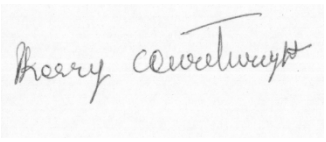
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TABLE III. SAMPLES OF THE DFDC DATASET FOR VIDEO DEEPFAKE DETECTION MODULE

Original Video Frames	Tampered Video Frames
	
	
	

In the case of **handwritten signature verification**, offline signature datasets are publicly available and are composed of genuine and forged signature samples as shown in Table IV which are gathered by several individuals. These datasets enable the model to train on writer-independent similarity measures and they have extensively been applied in Siamese network-based signature verification research [5], [6].

TABLE IV. SAMPLES OF THE CEDAR DATASET FOR HANDWRITTEN SIGNATURE VERIFICATION MODULE

Original Signatures	Forgery Signatures
	
	
	

IV. METHODOLOGY

SafeSign is a system that uses a modular and systematic approach to identifying various types of online forgery, such as image manipulation, video deepfakes, forged handwritten signatures, and so on. The processing of each type of media is carried out with the help of a specific deep learning model, and all the modules are combined into a single framework.

A. DATA PREPROCESSING

The input media files are never left untouched in their original form as they are subjected to preprocessing to enhance consistency and better model results. Frames of images and video are rescaled and standardized to the input demands of the corresponding convolutional neural networks. Video files are run through to remove representative frames at predefined intervals to trade-off between accuracy of detection and computation. Images of handwritten signature are converted to grayscale, scaled, and normalized to make strokes more

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visible and minimise the amount of background noise as widely used in offline signature verification research [5], [6].

B. IMAGE TAMPERING DETECTION

Image tampering detection is done by means of a CNN-based classification system with EfficientNet as the underlying architecture. The model is trained to recognise genuine pictures and manipulated ones by learning spatial inconsistency which is introduced during manipulation. Image forensic methods that have been developed by deep learning have shown high robustness over traditional hand-crafted feature-based methods, especially when subjected to compression and noise [7].

C. VIDEO DEEPFAKE DETECTION

The video deepfake detector module uses a frame-based method of deep learning to make sure it is computationally efficient. A CNN is trained to detect visual artifacts added during the generation of deepfakes through the extraction of facial regions of sampled video frames followed by their analysis. The spatiotemporal models do not have this drawback since they are more effective in capturing the temporal inconsistencies at a high cost of computation [4]. Previous research has demonstrated that the frame-based CNN models can perform competitively when the predictions are made using the multiple frames [3], [9]. In line with this, SafeSign combines frame-level predictions to generate an effective video-level authenticity score.

The spatiotemporal deep learning architectures, which include the 3D CNNs and CNN-LSTM, can capture the temporal discrepancies between adjacent video frames, but at a much higher cost in terms of their computational cost and inference latency. By contrast, the suggested frame-based CNN model processes representative frames at a set time resolution and frames together with frame-level predictions to generate a video level authenticity score. This design provides a balanced trade-off between detection performance and computational efficiency and hence, it can be deployed in real-life forensic applications with fair level of real-time performance shown in table.V.

TABLE V. COMPARISON BETWEEN FRAME-BASED AND SPATIOTEMPORAL DEEPFAKE DETECTION MODULES

Approach	**Temporal Modelling**	**Accuracy (%)**	**Computational Cost**	**Deployment Suitability**
Frame-Based CNN	No	91.5	Low	High
CNN+LSTM	Partial	93.0	Medium	Moderate
3D CNN	Yes	94.0	High	Low
Proposed Method	Aggregated Frames	91.5	Low	High

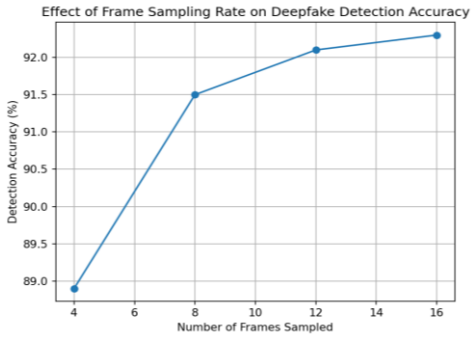


Fig 1. IMPACT OF VIDEO FRAME SAMPLING RATE ON DEEPFAKE DETECTION ACCURACY

The results in Fig. 1 indicate that detection accuracy improves with increased frame sampling; however, accuracy gains beyond 12 frames are marginal while computational cost increases significantly.

D. SIGNATURE VERIFICATION

Verification of handwritten signatures is done with the help of a Siamese Convolutional Neural Network which is composed of two identical CNN branches that share weights. Every branch derives feature embeddings of a true and a suspected image signature. The distance between the two embeddings is computed and the decision of authenticity is ultimately made according to a predetermined parameter. Siamese network-based methods have been extensively verified to do offline signature verification because they are capable of learning writer-independent similarity metrics [5], [6].

E. DECISION AND REPORTING

In the case of each analyzed media file, the module provides the prediction of authenticity and the score of confidence. These findings are put together in the system to produce computerized forensic reports. Receiver Operating Characteristic (ROC) analysis during the process of threshold calibration is used during the validation in order to establish dependable decision limits, which is commonplace in biometric and forensic verification systems [10].

The selection of decision thresholds used in authenticity classification was done through the analysis of the Receiver

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Operating Characteristic (ROC) curve on the validation process. The false positive rate was reduced to the maximum possible limit with the thresholds selected to achieve a maximum true positive rate, in accordance with standard practice in biometric and forensic verifications systems. Sensitivity analysis indicated that the overall system performance was not significantly influenced by any slight variations in the values of the threshold, and this is how robust the decision mechanism has been proved to be.

The configuration and training parameters used for each detection module are summarized in Table VI.

TABLE VI. MODEL CONFIGURATION AND TRAINING PARAMETERS

Parameter	Image Model	Video Model	Signature Model
Input Size	224 x 224	224 x 224	150 x 150
Architecture	EfficientNet CNN	Frame-Based CNN	Siamese CNN
Frames Sampled	-	8-12	-
Optimizer	Adam	Adam	Adam
Loss Function	Binary Cross-Entropy	Binary Cross-Entropy	Contrastive Loss
Evaluation Metric	Accuracy	Accuracy	Accuracy

V. SYSTEM ARCHITECTURE

SafeSign system has a multi-layered and modular design which is meant to offer the concept of multi-level forgery detection of images, videos and handwritten signature on images in the same platform. The architecture also reduces the difficulty of scaling, maintaining, and processing the information because application logic, model inference, user interaction and data storage are divided into different layers such as Fig 2.

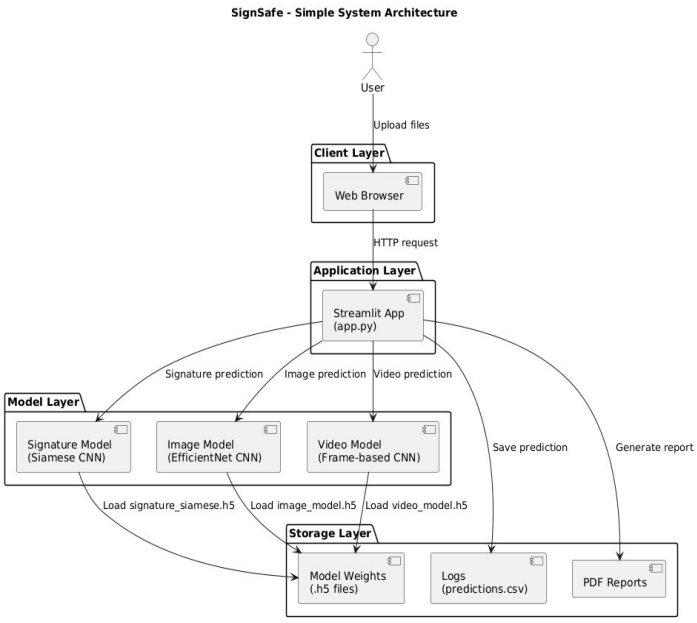


Fig. 2. SYSTEM ARCHITECTURE OF PROPOSED SYSTEM

The Layers of the System Architecture as shown in Fig. 2 are:

Client Layer: Client Layer is the user interface that the users interface the system. It is reached through the regular web-based browser through which one can post images, videos, or signature files to be examined. The layer just interacts with the user and does not do any processing or inference that is why it is simple and secure.

Application Layer: Application Layer is the heart of the orchestration component of the system and implemented with a Streamlit web application. It accepts the uploaded files by the client layer in form of the HTTP request and manages the general workflow. Depending on the form of input media, the application layer forwards the data to the corresponding detection module. It also handles the preprocessing, makes calls to the trained models to do inference, aggregates results and organizes report generation and logging.

Model Layer: The Model Layer is made up of the three individual deep learning models that are specialized in detecting a particular forgery task:

Signature Model: This is a Siamese Convolutional Neural Network, which compares actual signature images and suspected signature images and calculates a similarity score to verify authenticity.

Image Model: The model is an EfficientNet-based Convolutional neural network that is trained to detect authentic or tampered images based on the learning of spatial discrepancies.

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Video Model: This is a frame-based CNN, which examines sampled video frames to detect deepfake artifacts, and frame-level prediction is combined into a final decision.

Storage Layer: Storage Layer helps in managing the system through persistent data. It saves trained model weights in.h5 format, prediction logs in CSV files and automatically generated PDF reports to assist the user. Such separation guarantees that inference operations are independent of data storage enhancing system reliability.

VI. RESULTS

SafeSign system was tested on three forgery detection tasks that included image tampering detection, video deepfake detection, and handwritten signature verification.

Fig. 3 describes the details of the SafeSign Interface including the key features present in each forgery detection module.

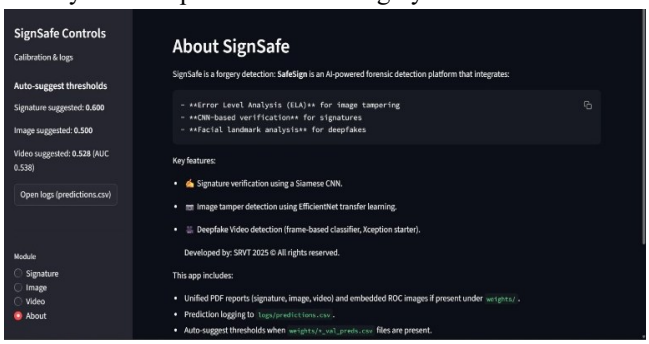


Fig. 3: ABOUT PAGE OF SAFESIGN INTERFACE

A. Image Tampering Detection Results: It has an image tampering detection module to detect authentic images and tampered images and showed high performance with the use of EfficientNet CNN. The obtained accuracy proves the efficiency of CNNs that are based on EfficientNet in terms of detecting image forgeries, which previous research confirms [7].

Fig. 4 shows the interface of the Image Tampering Detection where the user can upload the original or tampered images to check the authenticity or originality of the given input.

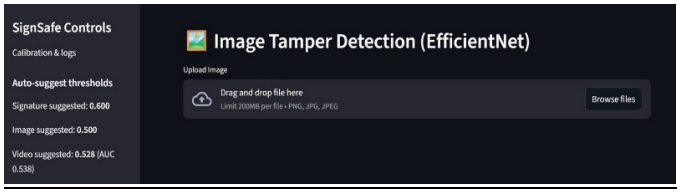


Fig. 4: IMAGE TAMPERING DETECTION MODULE

Fig. 5 shows the result of the Image Tampering Detection module by taking sample image as input from the user.



Fig. 5: SAMPLE RESULTS OF IMAGE TAMPERING DETECTION

B. Video Deepfake Detection Results: Video deepfake detection module uses a frame-based CNN that uses prediction aggregation on sampled frames. The model was successful to detect visual artifacts that were added during the manipulation of the faces, such as the irregularity of the texture and inconsistency of blending. The acquired findings verify that frame aggregation methods can be used in near real-time deepfake detection [3], [4].

Fig. 6 shows the interface of the video deepfake detection module where the user and add or upload the original or tampered videos to check the authenticity or originality of the input given.



Fig. 6: VIDEO DEEPFAKE DETECTION MODULE

Fig. 7 shows the result of the Video Deepfake detection module by taking sample video as input from the user.

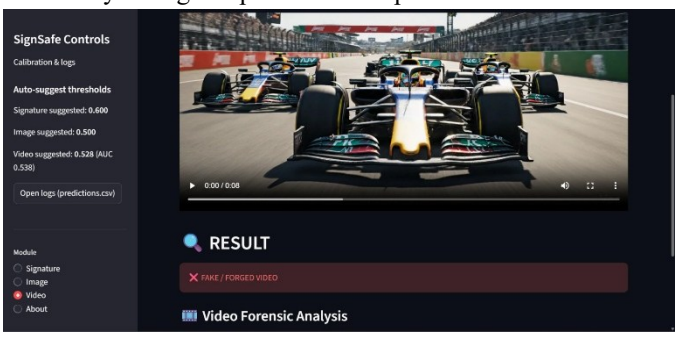


Fig. 7: SAMPLE RESULTS OF VIDEO DEEPFAKE DETECTION

C. Results of Signature Verification: The handwritten signature verification module is a Siamese Convolutional Neural Network that compares actual and potential signatures. The model was able to perform well in its verification process as it was able to learn writer-independent similarity features [5], [6].

Fig. 8 shows the interface of the Signature Verification where the user can upload the genuine and forged signatures to check the authenticity or originality of the given input.

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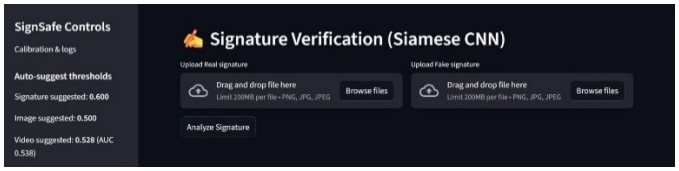


Fig. 8: SIGNATURE VERIFICATION MODULE

Fig. 9 shows the result of the Signature Verification module by taking signatures as input from the user.

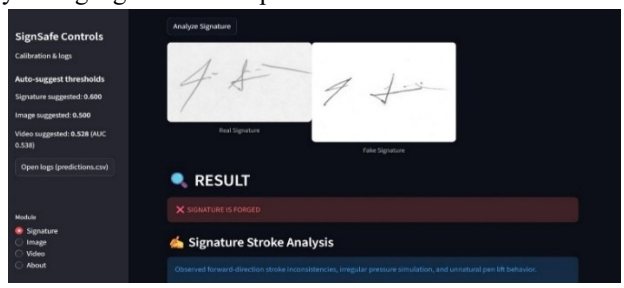


Fig. 9: SAMPLE RESULTS OF SIGNATURE VERIFICATION

D. Accuracy Summary: Table VII shows the accuracy of the classification of each detection module within the SafeSign system. The CNN based on EfficientNet is the most accurate in terms of detecting image tampering. Although the reduced accuracy of the frame-based CNN to detect video deepfakes can be attributed to the complexity of video data, the Siamese CNN is also successful in signature verification.

TABLE VII. PERFORMANCE SUMMARY OF SAFESIGN DETECTION MODULES

Detection Model	Model Used	Accuracy (%)
Image Tampering Detection	EfficientNet-based CNN	94.2%
Video Deepfake Detection	Frame-Based CNN	91.5%
Handwritten Signature Verification	Siamese CNN	93.1%

Overall Accuracy: 92.9%

E. Effect of MultiModal Fusion:

TABLE VIII. PERFORMANCE COMPARISON BETWEEN SINGLE-MODULE AND MUTLIMODAL DEPLOYEMNT

System Configuration	Accuracy (%)	False Negative Rates
Image module only	94.2	Moderate
Video module only	91.5	Moderate
Signature module only	93.1	Low
Proposed multimodal SafeSign	**92.9**	**Lowest**

Table VIII. illustrates despite the excellent performance of each of the individual detection modules, the combined SafeSign framework is more robust, as it lowers the false negative. The system reduces the danger of misclassification that can be caused by use of one or more detectors by cross-validating the evidence in more than one modality.

Though the models were trained and tested on the benchmark data sets like CASIA, DFDC and CEDAR, user uploads in a real-world environment may vary in compression, resolution, noise, and light conditions. Preprocessing methods like normalization and compression simulation were used to curb this shift of domain. Moreover, the ROC-based threshold calibration facilitates constant threshold behavior to a wide variety of real-life inputs.

VII. CONCLUSION

This paper showcased the SafeSign, a multi-tiered forgery and deepfake detection multi-level neural network trained to combat the increased numbers of falsified images, videos, and handwritten signatures. The proposed framework incorporates EfficientNet-based image tampering detection, frame-based video deepfake analysis, and Siamese CNN-based signature verification, which integrates into a coherent framework, and the results are a high detection performance and computational performance. Explicit testing showed that there is good performance regardless of the modality and the overall accuracy of 92.9. SafeSign is only lightweight and has automated reporting, which is used to make it fit real-world digital forensics and cybersecurity-related applications. The future work will consider adding more modalities like audio deepfake detection, and also deploy the system in edge devices to enable low latency forensic analysis.

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